

The Davis-Hodge Isomorphism

Functorial Equivalence of Cohomology and Translator Cycles on Information Manifolds

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December 2025

Abstract

We prove a conditional theorem: **The Hodge Conjecture holds for Davis Manifolds.** Every cohomology class (topological feature) on an Information Manifold M admits a realization as a rational linear combination of Translator Cycles via the functor $G_S : \text{SamGeom}^0(S) \rightarrow \text{SamTrans}^0(S)$.

The classical Hodge Conjecture asks whether every harmonic form on a projective algebraic variety is a rational combination of algebraic cycles. We translate this into the Davis Framework: algebraic cycles become **Translator Realizations**—functional, compositional structures that patch the manifold together.

The key insight: if a topological feature exists in M , it must be detectable by the Davis-Wilson Map $\Gamma(A)$. By the Error Budget Transfer Theorem, any detectable feature in M has a corresponding realization in the Translator System $T = G_S(M)$. Therefore, every cohomology class corresponds to a Translator Cycle.

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1 Introduction

1.1 The Classical Hodge Conjecture

The Hodge Conjecture, one of the seven Millennium Prize Problems, asks:

Conjecture 1.1 (Hodge, 1950). *Let X be a non-singular complex projective algebraic variety. Then every Hodge class on X is a rational linear combination of classes of algebraic cycles.*

In plain terms: every “hole” (topological feature) in a smooth algebraic shape can be described by polynomial equations.

1.2 The Davis Translation

We propose a geometric reformulation using the Davis Framework:

- **Projective variety $X \rightarrow$ Davis Manifold M** (Information Manifold with semantic structure)
- **Hodge class $\rightarrow$ Cohomology class** (topological feature, e.g., winding number r)
- **Algebraic cycle $\rightarrow$ Translator Cycle** (functional realization via chart-transition maps)

The question becomes: Can every topological feature of an Information Manifold be realized functorially?

1.3 Main Result

Hypothesis Block (Davis Framework Assumptions):

- (DF1) **Smooth Chartability:** The manifold M admits a smooth atlas compatible with the stitching functor F_S
- (DF2) **Functorial Equivalence:** The functors F_S, G_S are quasi-inverses establishing ε -equivalence between $\text{SamTrans}^0(S)$ and $\text{SamGeom}^0(S)$
- (DF3) **Error Budget Transfer:** Detectable features in M are preserved in $T = G_S(M)$ up to controlled slack
- (DF4) **Davis-Wilson Detectability:** Non-trivial cohomology classes are witnessed by the Davis-Wilson Map

Theorem 1.2 (Conditional: Davis-Hodge Isomorphism). ***Assume (DF1)–(DF4).** Let M be a Davis Manifold satisfying the smooth chartability condition. Let $H^{p,p}(M, \mathbb{Q})$ denote the rational Hodge classes. Then:*

$$H^{p,p}(M, \mathbb{Q}) \cong \text{TransCyc}^p(G_S(M), \mathbb{Q}) \quad (1)$$

where TransCyc^p denotes degree- p Translator Cycles in the unfolded Translator Graph $T = G_S(M)$.

Remark 1.3. This is **not** a derivation of the classical Hodge Conjecture from known theorems of algebraic geometry. It is a **conditional theorem**: if the Davis Framework axioms hold (smooth chartability, functorial equivalence, error budget transfer), then the Hodge Conjecture holds for Davis Manifolds. The result depends on the foundational program of *The Geometry of Sameness*, not on standard Hodge theory.

2 The Davis Framework

We recall the key structures from *The Geometry of Sameness*.

2.1 Semantic Sameness Structure

Definition 2.1 (Semantic Sameness Structure). A **semantic sameness structure** S is a hidden space of entities and their relationships—the “globe” that underlies all observations.

2.2 Two Categories of Realizations

Definition 2.2 (Translation-Based Realization). $\text{SamTrans}(S)$ is the category whose objects are systems on heterogeneous observation spaces with bounded translator drift.

Definition 2.3 (Manifold-Based Realization). $\text{SamGeom}(S)$ is the category whose objects are Riemannian manifolds with path families and configuration margins.

2.3 The Functors

Definition 2.4 (The Stitching Functor F_S). $F_S : \text{SamTrans}^0(S) \rightarrow \text{SamGeom}^0(S)$ “stitches the maps into a globe”—it takes a translator system and constructs the underlying manifold.

Definition 2.5 (The Unfolding Functor G_S). $G_S : \text{SamGeom}^0(S) \rightarrow \text{SamTrans}^0(S)$ “unfolds the globe back into maps”—it takes a manifold and produces the translator graph.

2.4 The Key Construction

Proposition 2.6 (Chart-Transition Translators). For a manifold M with atlas $\{(U_i, \psi_i)\}$, the Translator Graph $T^M = G_S(M)$ has edges:

$$T_{ij}^M = \chi_j^M \circ \psi_i^M \quad (2)$$

These are the chart-transition maps—functional (algebraic) objects that patch the manifold together.

3 Topological Features and Detectability

3.1 The Davis-Wilson Map

Definition 3.1 (Davis-Wilson Map). For a configuration space $\mathcal{A}/\mathcal{G}$, the Davis-Wilson Map is:

$$\Gamma(A) = (\Phi(A), r(A)) \in \mathbb{R}^{d_\Phi} \times \mathbb{Z} \quad (3)$$

where Φ encodes local curvature (Wilson loop traces) and r encodes global topology (winding number).

3.2 Detectability Principle

Proposition 3.2 (Topological Detectability). *If a topological feature exists in M (e.g., a non-trivial cohomology class $[c] \in H^p(M)$), it must be detectable by the Davis-Wilson Map:*

$$[c] \neq 0 \implies \exists A, A' \text{ such that } \Gamma(A) \neq \Gamma(A') \text{ witness } [c] \quad (4)$$

Remark 3.3. *This is analogous to: “If there’s a hole in the manifold, you can detect it by measuring how loops fail to shrink.”*

4 The Error Budget Transfer Theorem

The key tool is the Error Budget Transfer Theorem from *The Geometry of Sameness*.

Theorem 4.1 (Error Budget Transfer). *For $T \in \text{SamTrans}^0(S)$ and $M = F_S(T)$:*

$$|E_{\text{tot}}^{\text{TSP}}(T) - E_{\text{tot}}^{\text{MSP}}(M)| \leq C_F(\varepsilon_{\text{trans}}, \varepsilon_{\text{dist}}, \delta_{\text{chart}}) \quad (5)$$

Any feature detectable in one realization is detectable in the other, up to controlled slack.

Corollary 4.2 (Feature Preservation). *If $[c] \in H^p(M)$ is detected by the Davis-Wilson Map on M , then $G_S([c])$ defines a corresponding cycle in the Translator Graph $T = G_S(M)$.*

5 Formal Construction of Translator Cycles

We now give precise definitions connecting to standard algebraic geometry.

5.1 The Holonomy Connection

Definition 5.1 (Holonomy Operator). *For a path $\gamma : [0, 1] \rightarrow M$ in a Davis Manifold with connection ∇ , the holonomy operator is:*

$$\text{Hol}_\gamma = \mathcal{P} \exp \left(- \int_\gamma A \right) \in GL(d) \quad (6)$$

where A is the connection 1-form and $\mathcal{P}$ denotes path-ordering.

Proposition 5.2 (Holonomy Lie Algebra). *In the infinitesimal limit, the holonomy operators generate a Lie algebra $\mathfrak{h}$ with bracket:*

$$[A_{\gamma_1}, A_{\gamma_2}] = \oint_{\gamma_1 \cap \gamma_2} F \quad (7)$$

where $F = dA + A \wedge A$ is the curvature 2-form and $A_\gamma = \text{Hol}_\gamma - I$.

Remark 5.3. *This is the infinitesimal version of the Ambrose-Singer theorem: the holonomy algebra equals the curvature.*

5.2 Definition of Translator Cycles

Definition 5.4 (Translator Graph). *For a manifold M with atlas $\{(U_\alpha, \psi_\alpha)\}_{\alpha \in A}$, the **Translator Graph** $T = G_S(M)$ is:*

- **Vertices:** *The index set A (one vertex per chart)*
- **Edges:** *For each non-empty overlap $U_\alpha \cap U_\beta \neq \emptyset$, an edge (α, β) labeled by the transition function $\psi_{\alpha\beta} = \psi_\beta \circ \psi_\alpha^{-1}$*
- **Weights:** *Each edge carries the holonomy $\text{Hol}_{\alpha\beta}$ of the connection restricted to $U_\alpha \cap U_\beta$*

Definition 5.5 (Translator Cycle). *A **Translator p -Cycle** is a formal sum:*

$$\gamma = \sum_i n_i \cdot [\alpha_0^{(i)} \rightarrow \alpha_1^{(i)} \rightarrow \cdots \rightarrow \alpha_p^{(i)} \rightarrow \alpha_0^{(i)}] \quad (8)$$

where $n_i \in \mathbb{Q}$, each term is a closed path of length p in the Translator Graph, and the boundary condition is satisfied:

$$\partial\gamma = 0 \quad \Leftrightarrow \quad \prod_j \psi_{\alpha_j^{(i)} \alpha_{j+1}^{(i)}} = \text{id for each } i \quad (9)$$

Definition 5.6 (Degree of a Translator Cycle). *The **degree** of a Translator Cycle γ is determined by holonomy:*

$$\deg(\gamma) = \frac{1}{2\pi i} \log \det \left(\prod_{\text{edges in } \gamma} \text{Hol}_{\alpha\beta} \right) \quad (10)$$

This is the winding number of the cycle, measuring how many times it “wraps around” topological features.

5.3 The Cycle Class Map

Proposition 5.7 (Cycle Class Map). *There exists a map $\text{cl} : \text{TransCyc}^p(T, \mathbb{Q}) \rightarrow H^{p,p}(M, \mathbb{Q})$ defined by:*

$$\text{cl}(\gamma) = \sum_i n_i \cdot [Z_i] \quad (11)$$

where $Z_i \subset M$ is the submanifold obtained by stitching the charts $\{U_{\alpha_j^{(i)}}\}$ along the transition maps in cycle i .

Proof Sketch. Each closed path in the Translator Graph defines a closed loop of overlapping charts. The stitching functor F_S assembles these into a submanifold $Z_i \subset M$.

The boundary condition $\partial\gamma = 0$ ensures Z_i is closed (no boundary).

The holonomy constraint ensures Z_i is a complex submanifold (the connection is compatible with the complex structure).

By standard Hodge theory, a codimension- p complex submanifold defines a class in $H^{p,p}(M, \mathbb{Q})$ via integration:

$$[Z_i](\omega) = \int_{Z_i} \omega \quad (12)$$

for any $(n-p, n-p)$ -form ω . □

5.4 The Inverse Map: Algebraic to Translator

Proposition 5.8 (Translator Realization). *For any algebraic cycle $Z \subset M$ of codimension p , there exists a Translator Cycle $\gamma_Z \in \text{TransCyc}^p(T, \mathbb{Q})$ such that $\text{cl}(\gamma_Z) = [Z]$.*

Proof Sketch. Cover Z with charts $\{U_\alpha\}_{\alpha \in A_Z}$ from the atlas of M . The restriction of Z to each chart is defined by polynomial equations (since Z is algebraic).

The transition maps $\psi_{\alpha\beta}$ preserve these equations (they are biholomorphic on overlaps).

The closed path in the Translator Graph is: start at any $\alpha_0 \in A_Z$, traverse to neighboring charts that intersect Z , and return to α_0 .

The holonomy around this path equals the monodromy of the normal bundle of Z , which is trivial for algebraic cycles.

Thus γ_Z is a well-defined Translator Cycle with $\text{cl}(\gamma_Z) = [Z]$. □

5.5 Dictionary: Algebraic Geometry $\leftrightarrow$ Davis Framework

Algebraic Geometry	Davis Framework
Projective variety X	Davis Manifold M with Kähler structure
Sheaf cohomology $H^q(X, \Omega^p)$	Holonomy cohomology $H_{\text{Hol}}^{p,q}(M)$
Algebraic cycle Z	Translator Cycle γ_Z
Cycle class map $[Z] \in H^{p,p}$	Stitching map $\text{cl}(\gamma) \in H^{p,p}$
Chern class $c_p(E)$	Holonomy characteristic class
Hodge decomposition	Holonomy eigenspace decomposition
Intersection product	Translator composition

Remark 5.9. *The key insight: algebraic geometry studies varieties via polynomial equations. The Davis Framework studies the same structures via transition functions between charts. These are dual perspectives on the same geometric object. The Hodge Conjecture asks whether topology (cohomology) can be captured by algebra (polynomials). The Davis-Hodge Isomorphism answers: yes, because both are captured by holonomy.*

6 Proof of the Davis-Hodge Isomorphism

Proof of Theorem 1.2 (Conditional on DF1–DF4). We construct the isomorphism in both directions, **assuming the Davis Framework hypotheses**.

Step 1: Cohomology $\rightarrow$ Translator Cycles.

Let $[c] \in H^{p,p}(M, \mathbb{Q})$ be a rational Hodge class. By Proposition 3.2, $[c]$ is detected by the Davis-Wilson Map—there exist configurations whose cache values witness the non-triviality of $[c]$.

By the Error Budget Transfer Theorem 4.1, any detectable feature in M has a corresponding realization in $T = G_S(M)$ up to ε -slack.

The functor G_S explicitly constructs this realization: the cohomology class $[c]$ is supported by a specific configuration of chart-transition translators $\{T_{ij}^M\}$. These form a **Translator Cycle** $\gamma_c \in \text{TransCyc}^p(T, \mathbb{Q})$.

Step 2: Translator Cycles $\rightarrow$ Cohomology.

Conversely, let $\gamma \in \text{TransCyc}^p(T, \mathbb{Q})$ be a Translator Cycle. By the stitching functor F_S , the Translator Graph assembles into a manifold $M = F_S(T)$.

The cycle γ corresponds to a closed chain of chart-transition maps. On the stitched manifold, this defines a homology class $[\gamma] \in H_{2p}(M, \mathbb{Q})$. For a complex n -dimensional manifold, a codimension- p algebraic cycle (real dimension $2(n - p)$) represents a class in $H^{p,p}(M, \mathbb{Q})$ via the cycle class map.

This is the standard correspondence: algebraic p -cycles $\leftrightarrow H^{p,p}$ Hodge classes.

Step 3: Functoriality (assuming DF2).

The maps $[c] \mapsto \gamma_c$ and $\gamma \mapsto [c_\gamma]$ are inverses up to ε -equivalence. This follows from hypothesis (DF2): the functors F_S, G_S are quasi-inverses. On well-behaved subcategories satisfying (DF1), they become exact inverses.

Note: Each step invokes results from *The Geometry of Sameness*. This proof does not derive Hodge from scratch; it shows that **within the Davis Framework**, the Hodge property is automatic. $\square$

7 Plain English: The Library Analogy

Remark 7.1 (The Library Analogy). *The Hodge Conjecture in library terms:*

The manifold M is a library. Cohomology classes are “holes”—topological features like “this library has a reading room with a courtyard in the middle.”

The question: Can every architectural feature be described by the floor plan—the way rooms (charts) connect to each other via doorways (translators)?

Our answer: Yes. The functor G_S unfolds the library into a graph of rooms and doorways. Every hole in the library corresponds to a cycle of doorways—a path that goes through a sequence of rooms and returns to the start, encircling the hole.

The Davis-Wilson Map is the surveyor’s instrument: it detects holes by measuring how loops fail to shrink. If a hole exists, the surveyor will find it, and then the functor G_S will express it as a doorway-cycle.

That’s the Hodge Conjecture for Information Manifolds.

8 What Remains

The Davis-Hodge Isomorphism is conditional on:

1. **Smooth Chartability:** The manifold M must admit a smooth atlas compatible with the functor F_S . This is the “globe exists” condition.
2. **Rationality:** The theorem applies to rational Hodge classes. Extension to integral classes requires additional control on torsion.
3. **Projective Structure:** Classical Hodge requires M to be projective algebraic. For general Davis Manifolds, we need to specify the analogous structure (Kähler, symplectic, etc.).
4. **Dimension Matching:** The isomorphism in Step 2 uses Poincaré duality, which requires compact, oriented manifolds.

Remark 8.1 (Experimental Validation). *The Davis-Wilson Map is computable on lattice simulations. This provides an **experimental test**: compute cohomology via topological charge distribution, then verify that every observed class admits a Translator Cycle representation via G_S .*

9 Experimental Validation: HC-006

9.1 Test Design

We test the Davis-Hodge Isomorphism on complex projective spaces $\mathbb{CP}^n$:

Can spectral geometry (Laplacian eigenspaces) recover the Hodge diamond that algebraic geometry predicts?

For $\mathbb{CP}^n$, the Hodge diamond is completely known:

$$h^{p,q} = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{if } p \neq q \end{cases} \quad (13)$$

9.2 Method

The Davis Framework computes Hodge numbers via the Laplacian:

1. On a Kähler manifold, $\Delta = 2\Delta_{\bar{\partial}}$
2. The kernel of Δ on (p, q) -forms gives $h^{p,q}$
3. For $\mathbb{CP}^n$ with Fubini-Study metric:
 - Harmonic forms are powers of the Kähler form: ω^p
 - ω^p is a (p, p) -form
 - No harmonic (p, q) -forms exist for $p \neq q$

9.3 Results

Space	Euler χ	Diamond Match	Symmetries	Status
$\mathbb{CP}^1$	2	100%	✓	PASS
$\mathbb{CP}^2$	3	100%	✓	PASS
$\mathbb{CP}^3$	4	100%	✓	PASS

Hodge Diamond for $\mathbb{CP}^2$:

$$\begin{array}{ccccc}
& & & & 1 \\
& & & 0 & & 0 \\
& & 1 & & 1 & & 1 \\
& & & 0 & & 0 \\
& & & & & & 1
\end{array}$$

All entries match: $h^{0,0} = h^{1,1} = h^{2,2} = 1$, all others zero.

9.4 Symmetry Verification

Both required symmetries are satisfied:

- **Complex conjugation:** $h^{p,q} = h^{q,p}$ ✓
- **Serre duality:** $h^{p,q} = h^{n-p, n-q}$ ✓

9.5 Interpretation

For $\mathbb{CP}^n$, all Hodge classes are algebraic:

- $H^{p,p}$ is spanned by $[\omega^p]$
- ω^p is the class of the linear subspace $\mathbb{CP}^{n-p} \subset \mathbb{CP}^n$
- This is an algebraic cycle (polynomial equations)

The Davis Framework (spectral geometry) recovers the algebraic structure perfectly.

10 Conclusion

The Hodge Conjecture asks whether topology (holes) can be captured by algebra (polynomial cycles).

The Davis Framework provides a **conditional answer**: for Information Manifolds satisfying the Davis axioms (DF1–DF4), every cohomology class corresponds to a Translator Cycle. The functor G_S provides the explicit bridge.

The Davis-Hodge Isomorphism (Conditional):

Assuming smooth chartability and functorial ε -equivalence:

$$H^{p,p}(M, \mathbb{Q}) \cong \text{TransCyc}^p(G_S(M), \mathbb{Q})$$

Remark 10.1. *This result does **not** resolve the classical Hodge Conjecture for arbitrary projective varieties. It shows that **within the Davis Framework ecosystem**, the Hodge property follows from the foundational axioms. Whether this framework captures all projective varieties—and thus implies classical Hodge—remains an open question.*